

Question 4 — External Interrupts

Register Configuration

Two push buttons are connected to the external interrupt pins:

- Button 1 → INT0 / D2
- Button 2 → INT1 / D3

In the register-level implementation, the EICRA register is configured so both interrupts are triggered on the falling edge. The EIMSK register is then used to enable INT0 and INT1.

The interrupt service routines are:

- INT0: increases the pulse counter.
- INT1: resets the pulse counter to zero.

The ISR only updates counters or flags. Serial printing is handled later in the main loop.

Timing Calculations

A debounce time of at least 50 ms is used.

This means that after one valid interrupt is accepted, any additional interrupt that happens within the next 50 ms is ignored. This prevents one button press from being counted several times because of mechanical bouncing.

The timing is checked using `millis()` without using `delay()`.

System Behavior

When Button 1 is pressed, the pulse counter increases by one.

When Button 2 is pressed, the counter is reset to zero.

The updated counter value is displayed on the LCD, and the program also sends an event message to the Serial Monitor.

The register-level version performs the same operation as `attachInterrupt()`, but the interrupt configuration is done directly using AVR registers.